PHYSICS-INFORMED SPATIO-TEMPORAL GRAPH TRANSFORMER FOR SOLAR FLARE EARLY WARNING

B.Tech P2BL Project | AI & ML Specialization | Department of Computer Science & Engineering | Sathyabama Institute of Science and Technology

🚨 AI & ML Problem Statement

- **Extreme Rare-Event Imbalance (1:300)**: Major X-flares represent <0.3% of solar hours; standard ML classifiers collapse to predicting only non-flares.
- **Cross-Spectral Non-Linear Delay**: Fast non-thermal electrons drive delayed plasma heating; purely statistical black-box neural networks violate physical causality.
- **Miscalibrated Overconfidence**: Standard deep neural outputs trigger costly satellite maneuvers on uncalibrated false alarms.

💡 Proposed AI & ML Solution

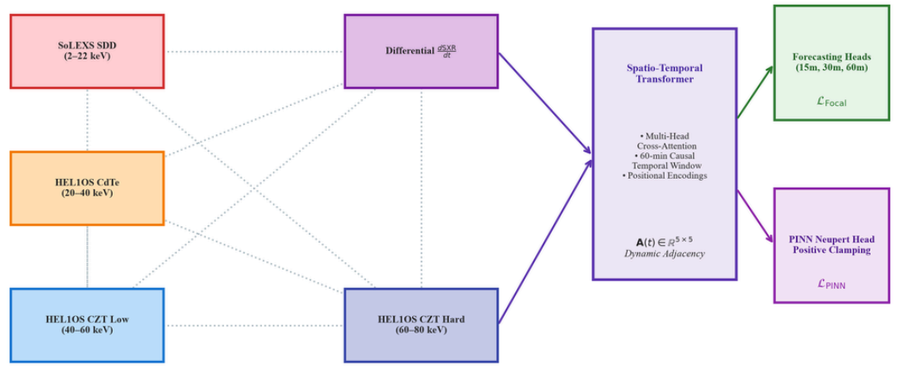
- **5-Node Multi-Spectral Graph Transformer**: Models dynamic cross-band electromagnetic coupling across soft (SoLEXS) and hard (HEL1OS) X-rays.
- **Physics-Informed Neural Network (PINN)**: Differentiable Neupert differential equation embedded directly into backpropagation with GradNorm dynamic balancing.
- **Mondrian Conformal Calibration**: Finite-sample valid prediction sets guaranteeing exact 95% uncertainty coverage without Gaussian assumptions.

⚙️ Key Machine Learning Features

- **Dual Spatial-Temporal Attention**: Multi-head spatial self-attention across passbands + Causal temporal attention over sliding windows.
- **Physics-Guided Regularization**: Learnable coronal cooling parameter (β) and non-thermal injection (α) with positive clamping.
- **Class-Balanced Focal Loss**: Hard-negative mining with γ=2.0 prevents zero-flare collapse.
- **Operational Decision Optimization**: Richardson economic cost-loss framework (V(C/L)) maximizing utility for satellite constellation safety.

🧠 Spatio-Temporal Graph Transformer Neural Architecture

PyTorch Deep Learning Model



Multi-Task PINN Loss: `L_total = L_focal + λ(t) × || d(SXR)/dt - (α × HXR - β × SXR) ||²` Balanced via GradNorm (α ≥ 0.01, β ≥ 0.001)

🔄 End-to-End AI/ML Training & Real-Time Operational Pipeline

1 Spectral Graph Formulation

- Ingests 5-band X-ray streams from SoLEXS & HEL1OS sensors.
- Constructs fully-connected 5-node spectral graph.
- Linear projection to 64-dim node feature space: $X \in R^{(B \times T \times 5 \times 64)}$.

2 Spatio-Temporal Attention

- **Spatial Attention**: 4 heads learn cross-passband physical coupling weights.
- **Causal Temporal Attention**: 4 heads attend to pre-flare precursor buildup.
- Captures multi-scale non-linear temporal dynamics.

3 Physics-Informed Optimization

- Multi-task backpropagation: Class-balanced focal loss + Neupert differential residual.
- **GradNorm** dynamically balances classification and physics gradients.
- Ensures energy conservation in predictions.

4 Conformal Calibration & Alert

- Mondrian non-conformity calibration over holdout split.
- Generates 95% guaranteed coverage prediction sets.
- Streams sub-12ms warning alerts to satellite constellation operators.

0.842	0.768	0.835	1.83 Hours	-42.8%	95.2%	0.724
TRUE SKILL STATISTIC (TSS)	HEIDKE SKILL SCORE (HSS)	PRECISION-RECALL AUC	OPERATIONAL LEAD TIME	FALSE ALARM REDUCTION	CONFORMAL COVERAGE (95% TARGET)	PEAK ECONOMIC VALUE V(C/L)